Check for updates

Short communication

IJMEE

International Journal of Mechanical Engineering Education
1–9
© The Author(s) 2026
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/03064190261418523
journals.sagepub.com/home/ijj
Sage

Reflections on AI in mechanical engineering education

Thomas J. Impelluso[1]

Abstract

Most current applications of AI in mechanical engineering emphasize ChatGPT-like tools for summarization or explanation, often overlooking AI's potential to guide both learning and research through structured navigation of complex problem spaces. This paper presents personal reflections on how AI can transform engineering pedagogy by fostering deeper learning, guiding students to actively construct understanding, and progressively supporting their development from guided exploration to independent reasoning. This shifts the emphasis from procedural mechanics (removing steps, sequencing tasks) to the cognitive journey of the learner, highlighting engagement, understanding, and independent thinking. By leveraging principles from Cognitive Load Theory—managing intrinsic, extraneous, and germane load—AI can dynamically tailor guidance to learners' evolving understanding. While this requires additional effort from the instructor, the payoff is increased engagement. The same guiding principles extend naturally to research and computational workflows. Rather than replacing research or numerical solvers, AI can operate within established frameworks—such as finite element methods and rigid multi-body dynamics—to identify physically meaningful starting points, navigate families of related solutions, and accelerate analysis while preserving theoretical integrity. In both educational and research contexts, AI functions not as a black-box solver but as a navigator of structured solution spaces. These reflections are intended as conceptual insights and practical proposals rather than a comprehensive literature review. In an era of rapid technological change, this approach—using AI as a

[1]Professor, Department of Mechanical and Marine Engineering, Western Norway University of Applied Science, Bergen, Norway

Corresponding author:
Thomas J. Impelluso, Department of Mechanical and Marine Engineering, Western Norway University of Applied Science, Bergen, Norway.
Email: Thomas.J.Impelluso@hvl.no

Data availability statement included at the end of the article

guide for both learning and computation—offers a coherent framework for deeper engagement, future research, and responsible integration into engineering education and practice, while formal assessment remains in abeyance.

Keywords
Artificial intelligence, mechanical engineering education, guided learning, cognitive load theory, multibody dynamics, finite element method, moving frame method, scaffolded learning

Scope and perspective

This paper presents personal reflections on the role of AI in mechanical engineering education and research. It is not intended as a conventional research article or a comprehensive literature review, but rather as a conceptual exploration of the author's perspectives and proposals. Given the rapidly evolving nature of AI in engineering education, preprints and arXiv publications are cited to provide timely examples of emerging research that may not yet appear in traditional archival journals but are shaping current practice. While the pedagogical and research components may initially appear distinct, they are intentionally connected and arise from the same premise: AI is most effective when it navigates structured problem spaces rather than replacing classical reasoning. In education, AI has been leveraged to guide students through scaffolded solution paths informed by Cognitive Load Theory, while in research and computation, the same principles allow AI to navigate families of physically meaningful solutions within established frameworks such as multibody dynamics and finite element methods. In this sense, the research component builds directly on the educational one, illustrating how guided learning principles scale naturally into professional engineering workflows.

Context of AI use in engineering education

Recent studies suggest that many current applications of AI in engineering education remain based in ChatGPT tools, superficial, often emphasizing shortcuts, text-based explanations, or automated solution generation rather than fostering deep learning, and are often used without much guidance. For example, Lund et al.[1] highlight ethical concerns and potential academic integrity issues that arise when generative AI is used without structured guidance, something this work hopes to avoid. Krause et al.[2] examine how AI affects student skills and instructor roles, showing that it can reshape teaching dynamics and engagement. Krupp et al.[3] document the negative consequences of unguided ChatGPT-assisted problem-solving in physics education, including over-reliance on AI outputs.

Similarly, Garousi et al.[4] propose a causal model to encourage responsible AI use in software engineering education, identifying key factors that influence ethical and effective AI integration and suggesting interventions to mitigate potential risks. Wecks et al.[5] analyze generative AI's influence on exam performance, while Pitts et al.[6] provide insights into student perceptions of AI's benefits and risks, emphasizing the need for balanced and informed integration to support learning without undermining critical thinking.

These findings underscore the importance of intentionally designing AI integration in education. Without structured guidance, AI use can risk undermining student-initiated work, but when thoughtfully implemented, it can scaffold learning and enhance active engagement. Rather than functioning merely as a source of solutions or explanations, the author of this paper has framed AI as a scaffold, guiding students through structured problem-solving pathways and promoting active engagement. In this way, this paper aligns with the broader discourse on responsible AI use in engineering education and demonstrates how guided approaches can support conceptual understanding while avoiding superficial application.

This philosophy has directly shaped the pedagogy deployed in the Moving Frame Method, where AI is used to selectively guide students through solution steps and progressively build independent problem-solving skills.

The moving frame method

The Moving Frame Method (MFM) adopts Élie Cartan's concept of placing a moving frame on all moving bodies[7] and leverages the mathematics of Sophus Lie,[8] distilled into the simplicity of matrices. This establishes a consistent methodology in which the same notation applies to 2D and 3D systems, as well as to single- and multi-body systems. The pedagogical foundations of the method are presented in,[9,10] and it has been widely applied in research and engineering practice.

Numerous peer-reviewed archival journal papers have employed MFM, addressing topics such as Subsea ROVs,[11] Gyroscopic Ship Stability,[12] Snake Robots,[13] Floating Wind Turbines,[14] the Tennis Racket Flip,[15] and the Stewart Platform.[16] Many conference proceedings also feature applications of MFM, with undergraduate-authored papers indicated in italics, including Gyroscopic Wave Energy,[11] Dual Gyroscopic Wave Energy,[12] Ship Stability,[13,14] Collisions,[15] and Robotic Cars.[16,17] For additional examples, educational materials, and resources, see the dedicated website: http://movingframe.org. Prior work has shown that MFM qualitatively engages students through the use of 3D mathematics, preparing them to shift from game-based thinking toward rigorous problem solving.

By placing a moving frame on every body in a system, MFM explicitly reveals the underlying geometric and dynamical patterns across 2D and 3D, single- and multi-body systems. These patterns naturally define a structured solution space, which is highly amenable to AI-guided learning. AI can navigate this structure to selectively highlight conceptually essential steps, guide students through relationships between frames, and scaffold problem-solving in a manner analogous to how the method organizes physical motion. In this way, MFM not only clarifies mechanics but provides a natural substrate for AI to structure and sequence learning tasks, helping students progressively build independent understanding while maintaining coherence with classical theory.

Moreover, the structured relationships exposed by MFM mirror the logic used in research workflows, allowing AI to recognize patterns, suggest solution pathways, and provide plausible initial conditions for numerical integration or multi-body analysis. This dual applicability demonstrates how a conceptually rigorous method can simultaneously enhance pedagogy and research, reinforcing the broader principle that structured

solution spaces, whether cognitive or physical, amplify learning and problem-solving rather than replace it.

Learning and pedagogical applications

Traditionally, the author's solution sets presented fully worked problems, including governing equations, step-by-step manipulations, and final expressions. While this approach ensured clarity for post-submission study, it often encouraged passive reproduction rather than active problem-solving. By contrast, AI has enabled this author to facilitate guided exploration, progressively fading scaffolding to lead students toward independent derivation, leveraging Cognitive Load Theory (CLT) (Sweller[18]). The author has previously applied CLT explicitly in the design of a massively open online course for teaching computer coding to mechanical engineering, demonstrating measurable improvements in student learning and engagement (Impelluso[19]).

Cognitive Load Theory distinguishes three types of load on learning:

- Intrinsic load – the inherent difficulty of the content.
- Extraneous load – the way information is presented, including unnecessary steps or confusing instructions.
- Germane load – the mental effort devoted to building schemas and understanding patterns.

AI can analyze solution steps and identify those imposing the heaviest intrinsic load. In practice, the author begins with a complete solution and selectively removes steps. ChatGPT is then used to recommend which intermediate steps to remove, encouraging active problem-solving while avoiding cognitive overload.

AI can also sequence tasks to gradually increase difficulty, fading scaffolding at an appropriate pace so that germane load is maximized. For example, in a class where each major topic is organized into five tasks of five problems each, the first task may remove only 10% of intermediate steps, nudging students to engage actively while still providing a clear model. Later tasks may omit 20–40% of steps, and by the final tasks, students are expected to perform the entire derivation independently. AI can suggest step removal based on student performance patterns and cognitive load, maintaining an optimal balance between challenge and support.

In this framework, students are never abandoned; they are guided along structured paths. Problem sets become exploratory journeys rather than fully prescribed instructions. Prompts such as "here is what to do next" remain intact, while intermediate steps are selectively omitted. Students are encouraged to reason through derivations, detect patterns, and actively apply their understanding. In this way, the learning experience mirrors the research process: AI functions as a guide navigating structured spaces, whether these are physical solution manifolds in computation or cognitive solution paths in learning.

Creating the task with the use of AI

In the following illustrative example (details simplified for clarity), ChatGPT was provided with a section of a worked solution and asked to evaluate each step in terms of guided learning, using Cognitive Load Theory as a framework. ChatGPT analyzed each step and recommended whether it should be left intact, partially removed, or fully removed, providing reasoning based on intrinsic, extraneous, and germane load.

This exercise demonstrates how AI can interact with students in practice: it identifies steps that are conceptually essential, procedural steps that can be omitted, and steps that can be partially guided to encourage active reasoning. Importantly, this approach is adaptable to different stages of learning: early exercises retain more guidance for novices, intermediate exercises partially remove steps to promote schema development, and advanced exercises provide minimal guidance to foster independent problem-solving.

The example illustrates AI's potential to dynamically scaffold learning, guiding students progressively from structured exploration toward independent mastery.

Chat prompt used

"I will paste a full worked solution to a problem [and I did]. Please go through each step and provide an itemized list stating: (1) whether the step should be left, partially removed, or fully removed to guide student learning, (2) your reasoning for that choice, and (3) if relevant, how it relates to cognitive load theory (intrinsic, extraneous, germane). Focus on clarity and brevity for each item."

Step 1: Establish inertial and rotating frames at the origin

- **Recommendation:** Leave this in
- **Reason:** Essential for orientation and context. Provides the foundation for all subsequent reasoning. Removing it would increase **intrinsic load** by forcing students to infer reference frames themselves.

Step 2: Highlight axes conventions (1 axis right, 2 axis into page)

- **Recommendation:** Leave this in
- **Reason:** Crucial for correctly applying the right-hand rule. Supports **germane load** by reinforcing proper conceptual mapping.

Step 3: Obtain moving frame from inertial frame using a rotation matrix

- **Recommendation:** Remove partially
- **Reason:** Procedural algebra can be removed; letting students construct the rotation matrix themselves increases **germane load** and avoids unnecessary **extraneous load** from copying formulas.

Step 4: Invert rotation matrix to obtain inertial frame from moving frame

- **Recommendation:** Remove completely
- **Reason:** Standard formulaic step; leaving it would not add conceptual value. Encourages active retrieval and schema building.

Step 5: Take time derivative of moving frame (recognizing inertial frame is fixed)
- **Recommendation:** Leave this in
- **Reason:** Conceptually necessary. Students must understand why the inertial frame does not change over time, which is intrinsic to the problem.

Step 6: Insert inertial frame to express moving frame rate in moving frame coordinates
- **Recommendation:** Remove partially
- **Reason:** Multiplication and substitution are procedural. Letting students carry out the operation themselves strengthens **germane load** by focusing on frame relationships rather than algebra.

Step 7: Define omega-skew and simplify
- **Recommendation:** Remove completely
- **Reason:** Procedural step; derivation of omega-skew can be guided with hints but doesn't need full worked-out solution. Supports active learning.

Step 8: Take derivative of particle position vector (distance d constant)
- **Recommendation:** Remove partially
- **Reason:** Let students compute the derivative themselves; this builds understanding of how position transforms in a moving frame while keeping **intrinsic load manageable**.

Step 9: Set up forces and equate to mass × acceleration
- **Recommendation:** Leave this in
- **Reason:** Conceptually essential. Forces must be identified explicitly; removing them would confuse the connection between kinematics and dynamics.

Step 10: Apply boundary condition when tension reaches 2 N to solve for time
- **Recommendation:** Leave this in
- **Reason:** This is the key problem-solving step. Students must practice applying results to physical conditions, supporting **germane load** and providing closure.

Taken together, this stepwise guidance ensures that students are never lost in the solution process: they understand which steps are essential, which require active construction, and how the overall problem unfolds. By progressively revealing the solution while selectively withholding procedural details, AI scaffolds learning and provides students with a clear path forward, enabling them to navigate the problem space confidently and develop independent problem-solving skills.

Research and computational applications

At first glance, the pedagogical and research-oriented components of this paper may appear distinct. However, they are intentionally intertwined, stemming from the same premise: AI is most effective when it navigates structured solution spaces rather than replacing them. The guided learning approach in the classroom—where AI scaffolds student problem-solving using Cognitive Load Theory—is mirrored in professional workflows for research and computation.

In both education and research, progress depends on identifying meaningful structure, selecting appropriate representations, and efficiently traversing a space of plausible

solutions. Guided learning in the classroom reflects how expert engineers and researchers reason about complex systems.

Rather than discarding classical frameworks such as finite element theory, rigid body dynamics, or differential geometry, AI can navigate within them—linking small models to larger ones, refining coarse solutions, and reusing component knowledge to accelerate complex analyses while preserving theoretical integrity.

In finite element methods (FEM), AI can analyze canonical problems with coarse meshes, simple geometries, and standard loads, recognizing that these solutions already contain essential large-scale structure. These displacement fields form a structured "solution library," with each entry embodying smoothness, mode shapes, stiffness coupling, and boundary effects. For new problems, AI can identify the most relevant base solution and expand or refine it using multiresolution techniques such as wavelet transforms, producing physically meaningful fine-scale approximations suitable for iterative solvers. This approach reduces computational time while maintaining theoretical fidelity.

Similarly, the Moving Frame Method (MFM) expresses rigid-body motion from a geometry-first perspective, replacing a single inertial frame with multiple moving frames. Each link carries its own differential relationships, revealing the structure of motion rather than relying on ad hoc coordinate choices. Once link behaviors are encoded as canonical geometric patterns, AI can search, select, and assemble them into physically consistent trajectories for larger mechanisms. This enables AI to navigate motion, scale or combine solutions, and provide plausible initial conditions for numerical integration, reducing computational errors and improving stability.

The Moving Frame Method, by placing a moving frame on every body in a system, exposes the underlying geometric and dynamical patterns across 2D and 3D, single- and multi-body systems. These explicit patterns create a structured solution space that is highly amenable to AI-guided learning: AI can navigate this structured framework to selectively highlight essential steps, guide students through conceptual relationships, and scaffold problem-solving in a manner analogous to how the method organizes physical motion. In this way, MFM not only clarifies mechanics but provides a natural substrate for AI to structure and sequence learning tasks, helping students progressively build independent understanding while maintaining coherence with classical theory.

Both FEM and multibody dynamics illustrate a shared principle: engineering solutions exist in families. A beam bending under a distributed load relates to a plate bending under localized conditions, just as links in planar and spatial mechanisms share geometric structure. Recognizing these relationships allows AI to navigate solution spaces intelligently across scales and configurations, rather than generating physics from scratch.

In this framework, AI amplifies classical theory instead of replacing it. By detecting patterns, identifying representative solutions, and guiding computations within trusted mathematical and physical structures, AI functions as a navigator of structured solution spaces—mirroring the principles applied in pedagogy and providing a consistent methodology across education and research.

Finally, encouraging students to identify and explore similar problems on their own—while AI provides guidance—remains an important and instructive challenge in the learning process, demonstrating how AI can subtly guide without supplanting the learner's

reasoning. After each assignment, students are required to submit a brief statement on how the current problem is similar to or different from the previous one.

Conclusion

By integrating pedagogical heuristics with structured problem pathways, AI becomes a bridge: linking foundational concepts to more complex applications, preserving the integrity of classical methods, and extending students' reasoning capabilities. Similarly, in research and computation, AI can navigate structured solution spaces—whether in finite element methods or multibody dynamics—amplifying classical theory rather than replacing it.

The perspectives and examples presented here are intentionally conceptual. They illustrate practical ways AI can function as a navigator of structured solution spaces, offering a framework that can inspire experimentation, refinement, and adaptation by other educators and researchers. These ideas are starting points, not definitive prescriptions, and they invite the broader community to explore, implement, and evaluate their potential impact on engineering education and professional practice.

Framing AI as a guide through structured problem spaces offers a novel approach, unexamined to date, with implications for both engineering education and computational practice."

ORCID iD

Thomas J. Impelluso https://orcid.org/0000-0003-0457-5912

Funding

The author received no financial support for the research, authorship, and/or publication of this article.

Declaration of conflicting interests

The author declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data availability statement

Example text of a Data statement, as provided by the author.

References

1. Lund B, Mannuru NR, Teel ZA, et al. Student perceptions of AI assisted writing and academic integrity: Ethical concerns, academic misconduct, and use of generative AI in higher education. *AI in Education* 2025; 1: 2.
2. Krause S, Dalvi A and Zaidi SK. Generative AI in education: Student skills and lecturer roles. arXiv preprint arXiv:2504.19673. 2025. doi:10.48550/arXiv.2504.19673.
3. Krupp L, Steinert S, Kiefer Emmanouilidis M, et al. Unreflected acceptance—investigating the negative consequences of ChatGPT-assisted problem solving in physics education. arXiv preprint arXiv:2309.03087. 2023. Available from: https://arxiv.org/abs/2309.03087.

4. Garousi V, Jafarov Z, Movsumova A, et al. Encouraging students' responsible use of GenAI in software engineering education: A causal model and two institutional applications. arXiv preprint arXiv:2506.00682. 2025. doi:10.48550/arXiv.2506.00682.
5. Wecks JO, Voshaar J, Plate BJ, et al. Generative AI usage and exam performance. arXiv preprint arXiv:2404.19699. 2024. Available from: https://arxiv.org/abs/2404.19699.
6. Pitts G, Marcus V and Motamedi S. Student perspectives on the benefits and risks of AI in education. arXiv preprint arXiv:2505.02198. 2025. Available from: https://arxiv.org/abs/2505.02198.
7. Cartan E. *On Manifolds with an Affine Connection and the Theory of General Relativity. Trans. A. Magnon and A. Ashtekar*. Napoli, Italy: Bibliopolis, 1986.
8. Lie S. *Theory of Transformation Groups I: General Properties of Continuous Transformation Groups. Trans. contemporary edition*. Berlin: Springer Verlag, 2015, doi:10.1007/978-3-662-46211-9.
9. Impelluso TJ. The moving frame method in dynamics: Reforming a curriculum and assessment. *Int J Mech Eng Educ* 2017; 45: 1–34.
10. Impelluso TJ. The moving frame method in rigid body dynamics: Reflections on improving a discipline. *Int J Mech Eng Educ* 2025; 0. doi:10.1177/03064190251340176
11. Norbach A, Fjetland KB, Hestetun GV, et al. Gyroscopic wave energy generator for fish farms and rigs. In: Proc. ASME 2018 International Mechanical Engineering Congress and Exposition, Pittsburgh, PA, USA, Nov. 2018. Paper No. V06BT08A021. Available from: https://doi.org/10.1115/IMECE2018-86188.
12. Korsvik HB, Rognsvåg ES, Tomren TH, et al. Dual gyroscope wave energy converter. In: Proc. ASME 2019 International Mechanical Engineering Congress and Exposition, Salt Lake City, UT, USA, Nov. 2019. Paper No. V006T06A008. Available from: https://doi.org/10.1115/IMECE2019-10266.
13. Nyland J, Teigland H and Impelluso TJ. Use of the Moving Frame Method in Dynamics to Model Gyroscopic Control of Small Crafts at Sea: Theory — Part 1. IMECE2017-70108, V04AT05A059, 12 pages. 2018. https://doi.org/10.1115/IMECE2017-70108.
14. Teigland H, Flåten A, Lied M, et al. Use of the Moving Frame Method in Dynamics to Model Gyroscopic Control of Small Crafts at Sea: Experiment — Part 2. IMECE2017-70109, V04AT05A027. 2017. https://doi.org/10.1115/IMECE2017-70109.
15. Sværen T, Nygård B and Impelluso T. A Framework for Spatial 3D Collision Models: Theory and Validation. IMECE2021-72981. 2021. https://doi.org/10.1115/IMECE2021-72981.
16. Chen J, Kim E, Leppitsch C, et al. Part 1: Gyroscopic Control of Robotic Smart Vehicles Using SO(3). IMECE2023-113518, V008T09A042, 12 pages. 2024. https://doi.org/10.1115/IMECE2023-113518.
17. Rykkje T, Johnsen K, Hole PS, et al. Part 2: Gyroscopic Control of Robotic Smart Vehicles Using SE(3). IMECE2023-111506, V008T09A039, 11 pages. 2024. https://doi.org/10.1115/IMECE2023-111506.
18. Sweller J. Cognitive load during problem solving: Effects on learning. *Cogn Sci* 1988; 12: 257–285.
19. Impelluso TJ. Assessing cognitive load theory to improve student learning for mechanical engineers. *Am J Distance Educ* 2009; 23: 179–193.